Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Advanced Algorithms and Parallel Programming 1st part–June 11th, 2026

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

A. Describe the execution model of a PRAM computation. (1)

B. Discuss the possible conflicts that may arise during memory access in the PRAM model. (1)

C. Explain how shared memory is modeled in PRAM and how this affects complexity analysis. (1)

D. Discuss the scalability laws presented in class and their interpretation in terms of time and processors. (1)

Exercise n. 2

A. Describe amortized analysis and discuss one algorithm or data structure where this analysis is appropriate. (2)

B. Explain the role of Counting Sort inside Radix Sort. (1) **This question can be skipped if you got 1 or 3 points in the first challenge.**

C. Please explain what "Lower bound for comparison sorting" means and its implications. (1)

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Exercise n. 3

A. Please describe how the treap split works. (1)

B. Discuss Strassen's matrix multiplication algorithm and compare its complexity with the standard matrix multiplication algorithm. (1)

C. Describe the randomized primality testing algorithm presented in class, including its randomized-algorithm classification. (2)

Exercise n. 4

A. Detail the Longest Common Subsequence algorithm: pseudo code, space, and temporal complexity. (2)

B. Discuss self-organizing lists, including one heuristic presented in class and the method used to analyze its performance. (2) **This question can be skipped if you got 2 or 3 points in the first challenge.**